



DEALER BEST PRACTICE SERIES

Self Contained Bulk Oil Filtration Day Tank

Application Maintenance Site Management Component Rebuild Safety MARC Management

- Self Contained Bulk Oil Filtration Day Tank 0
- 1.0 Introduction 1
- 2.0 Best Practice Description..... 1
- 3.0 Implementation Steps 2
- 4.0 Benefits..... 3
- 5.0 Resources Required 3
- 6.0 Supporting Attachments / References 3
- 7.0 Related Best Practices 3
- 8.0 Acknowledgements..... 3

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

1.0 Introduction

It is important to fill newly rebuilt or repaired engines and powertrain components with clean oil. Dealers and customers are looking for multi-purpose oil storage tanks for their shops, which can also clean new oil. This system provides a consistent source of clean oil based on daily usage.

- Nearly all large shops have bulk oil tanks for various grades of oil. Unfortunately, most oil picks up particulate debris from the transportation and storage process before it reaches the end user.
- Filtering large quantities of oil requires large circulation/filtration units and must be operated continuously, often producing unpredictable results.
- In order to minimize contamination related failures of new rebuilt, or repaired components, it is important to minimize three sources of debris:
 - o Assembly debris in the component
 - o Break-in debris generated by the component
 - o Debris in new oil used in the component
- Many machine powertrain and hydraulic systems use electro-hydraulic valves to provide precise control of implement operation and clutch engagement. These valves require very clean oil to function properly. Cleaning bulk oil before it is put into the machine system reduces the opportunity for contamination-related failures.
- Providing a system of filtering only the oil that will be used in a specific time interval reduces filtration time as well as increasing fill oil quality.

Using clean bulk oil reduces early hour contamination failures and helps to improve reliability.

2.0 Best Practice Description

The self-contained bulk oil filtration day tank automatically provides clean oil for shop use. This unit draws oil from the much larger bulk storage tank and holds an adequate amount to supply the shop for a day's operation. The unit also filters the stored oil automatically through an integrated pump and filter system, which has a number of safeguards to assure proper operation. This automatic fill and filtration system provides the user the option of "off time" fill, manual override, programmable kidney loop duration and on demand delivery to point of use. The following are major features:

- Double walled tank to minimize spills and leakage. 100 or 280 gallon capacity.
- Single-walled tanks available for containment areas.
- 3 hp electric circulation pump provides a steady 25 gpm flow to the filters to maximize filtration efficiency.
- 3 micron or 5 micron series high efficiency filter elements with 30 psi filter bypass and electrical plugged filter indicator.
- 12 gpm 6:1 air operated supply pump with thermo relief supplies clean, filtered outlet oil to the oil delivery system.
- Secondary tank breather filter prevents dirty air from entering the outer tank chamber
- Desiccant breather filter provides clean, dry air to the inner primary tank.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Self Contained Bulk Oil Filtration Day Tank	DATE 2/17/2015	CHG NO 02	NUMBER 1106-2.14-1039

- Oil inlet to refill the primary tank has 6 micron high efficiency filters and automatic high level shut-off valve.
- Primary tank has a visual pop-up sight gage.
- Mechanical low level float switch prevents operation of the circulation pump if tank level is low.
- Electrical control panel contains the following features:
 - o 24 hour / 7 day timer for automatic operation
 - o Emergency stop button
 - o Pump on switch
 - o Start cycle light
 - o Low fluid level light
 - o Plugged filter light for each filter
 - o May be programmed for manual or automatic “off time” fill
- SOS sampling valve on filter inlet to measure cleanliness of supply oil.



Caterpillar automatic bulk oil filtration tank available from Cat® Service Tools



Bulk oil filters as an integral component of a dealer's tank farm in South America

3.0 Implementation Steps

1. Determine filtration needs for the particular shop
2. Purchase and install appropriate tanks

Information is available from the Caterpillar Service Tools Group.

Contact:

John Jordan

NACD/LACD Technical Sales Support

Capital and Shop Products

Ph: 309-636-1791

Cell: 309-253-1145

jordan_john_a@cat.com

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Self Contained Bulk Oil Filtration Day Tank

DATE
2/17/2015

CHG
NO
02

NUMBER
1106-2.14-1039

4.0 Benefits

- Using consistently clean oil reduces early hour contamination failures and improves reliability.
 - o Finning Chile utilizes two of the units for HYDO and Engine pre-fill oil applications achieving consistent 16/13 cleanliness levels.
 - o Wayne supply utilizes one of the units for engine pre-fill oil providing consistent 16/13 cleanliness levels.

5.0 Resources Required

- Investment Costs: \$12,000 - \$18,000 (depending on size and installation location).
- Facility: It is recommended this unit be installed in an enclosed location where temperatures remain between 15 – 32 °C (60 – 90 °F) to ensure consistent oil flow rate and minimize internal tank condensation.
- Tools: standard maintenance tools.
- Support equipment: A low-pressure delivery pump from the bulk tank to the day tank. This may be controlled by the day tank processor when fill is required.
- People: Daily monitoring of operation and filter condition is recommended.
- Training: SOS sampling and/or Particle Counter operation is recommended.

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

1106-2.10-1037 Bulk Fuel Storage & Handling

8.0 Acknowledgements

This Best Practice was written by:
 Richard Douglas
 Market Consultant
 Douglas_Richard_d@cat.com
 (309) 675-5699

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Self Contained Bulk Oil Filtration Day Tank	DATE 2/17/2015	CHG NO 02	NUMBER 1106-2.14-1039
---	-------------------	-----------------	--------------------------